

15 Catalytic Rate Laws

The reaction $2A \rightarrow B$ is carried out in the gas phase over a catalyst. The rate law is:

$$-r_A = \frac{kP_A^2}{(1 + K_A P_A + K_B P_B)^2}$$

Where $k = 1250 \frac{\text{mol}}{\text{atm}^2 \cdot \text{L} \cdot \text{s}}$, $K_A = 4 \text{ atm}^{-1}$, and $K_B = 13 \text{ atm}^{-1}$.

"Langmuir-Hinshelwood" type rate law -
comprises information about the free site coverage; the rate determining step

What is different about this rate law, compared to the ones we've been working with?

1. Rate given in terms of pressure instead of concentration

(recall for gases $C = P/RT$)

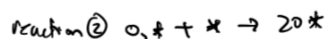
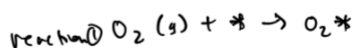
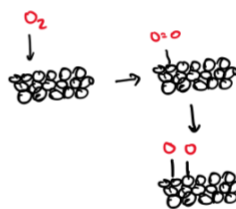
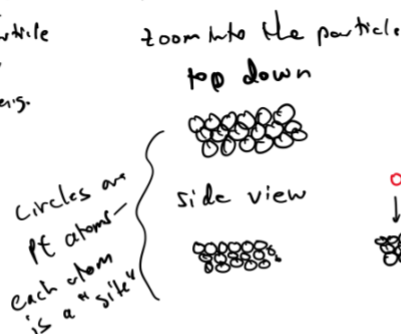
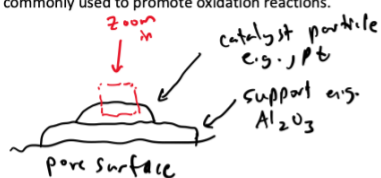
2. There is an equation in the denominator that is not a simple power law function of concentration or pressure

This arises because the reaction takes place on the catalyst surface, which has a finite # of sites to carry out the reaction

On a catalyst, a "site" is

a location or chemical identity on the catalyst surface where the rxn can take place

An example of a catalyst site is a Pt atom on a slab of Pt metal. For example, the Pt metal atom will easily dissociate the O_2 molecule, and hence Pt catalysts are commonly used to promote oxidation reactions.



where $*$ = catalyst site
*ed species are bound to a catalyst site

The denominator is a "coverage term". It tells us something about the fraction of free sites on the catalyst surface

What is coverage?? We will explore this in more detail later, but for now, let's pause and illustrate coverage: [links to powerpoint in OneDrive](#)

Similar to concentration. Tells the number of sites on the catalyst surface that are occupied by adsorbates (molecules/atoms). We are interested in the "free site" coverage because that's where the reaction can take place. In general coverage is denoted θ .

The free site coverage is denoted θ_*

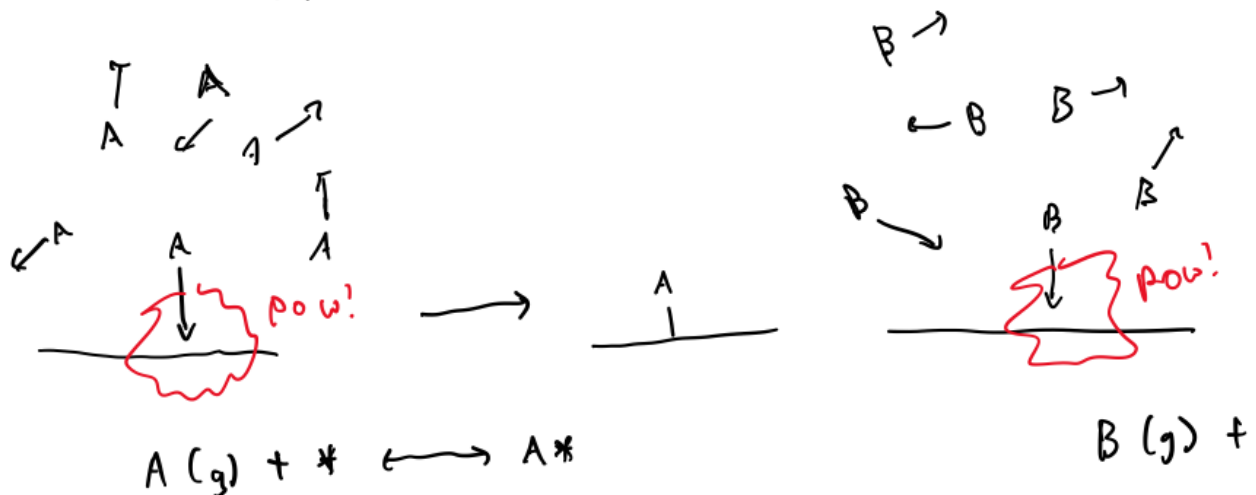
θ_* is the fraction of catalyst sites that don't contain adsorbates & hence are available to carry out the reaction

In our example, $\theta_* = 1 - \theta_A - \theta_B$ (in monolayer terms)

So, now we need expressions for θ_A and θ_B .

We get these by considering the adsorption of these species on the catalyst.

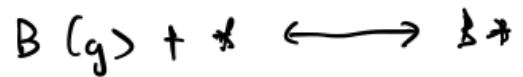
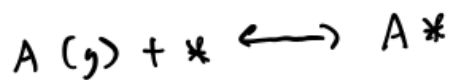
Adsorption occurs when a fluid phase molecule forms a bond with a solid surface



We assume that A (g) and B (g) are in equilibrium with the catalyst

$$0 = k_{f,A} P_A \theta_* - k_{r,A} \theta_A$$

$$0 = k_{f,B} P_B \theta_* - k_{r,B} \theta_B$$



forward reactions: adsorption, reverse rxns: desorption

Hence, we can write $K_A =$

and $K_B =$

$$K_A = \frac{\theta_A}{P_A \theta_*}$$

$$K_B = \frac{\theta_B}{P_B \theta_*}$$

$$\text{Hence, } \theta_* = 1 - \theta_A - \theta_B$$

$$\theta_A = K_A P_A \theta_*, \theta_B = K_B P_B \theta_*$$

$$\theta_* = 1 - K_A P_A \theta_* - K_B P_B \theta_* = \frac{1}{1 + K_A P_A + K_B P_B}$$

$$\text{Our rate law is } -r_A = \frac{k P_A^2}{(1 + K_A P_A + K_B P_B)^2} = K P_A^2 \theta_*^2$$

How do the coverages of A and B influence the rate? Why?

$\uparrow \theta_A$ and/or $\theta_B \downarrow -r_A$

A & B take up sites $\therefore$ hence inhibit the reaction